

$$x(0) = 1 \quad x(1) = 0 \text{ (from eq 2 \& 3)}$$

Where A and B are constants to be determined by the boundary condition. Some elementary algebra then leads to the exact solution of TPBVP.

$$u^*(t) = \frac{-2e}{1+e^2} \sinh(1-t)$$

$$x^*(t) = \frac{2e}{1+e^2} \cosh(1-t)$$

while the above example could be solved analytically, it is far less complex than that one can expect to encounter in practice. In fact, most practical optimal control problem cannot be solved analytically.

23.12 EXERCISES

Q1. The problem is to find the optimal flows of cake munching $C^* = \{C^*(t) : t \in [0, T]\}$ and of the size of the cake $W^* = \{W^*(t) : t \in [0, T]\}$ such that

$$\max_c \int_0^T \ln(c(t)) e^{-\rho t} dt \text{ subject to } W' = -C, t \in (0, T)$$

$$W(0) = \phi, W(T) = 0 \text{ where } \phi > 0 \text{ is given}$$

Ans. Our objective function is $\max_c \int_0^T e^{-\rho t} \ln(ct) dt$

$$s.t. = W' = -C$$

$$\text{Therefore } H^* = e^{\rho t} H(.) = \ln(ct) + \mu_t(-c_t)$$

The necessary condition for optimisation

$$\frac{\partial H^*}{\partial u_t} = 0 \Rightarrow \frac{\partial}{\partial u_t} \ln(ct) + \mu_t \frac{\partial}{\partial u_t} (-ct) = 0 \text{ where } u_t = c_t$$

$$\Rightarrow \frac{1}{c_t} + \mu_t(-1) \Rightarrow c_t = \frac{1}{\mu_t} \quad \dots (i)$$

$$\text{Setting } \frac{\partial H^*}{\partial W_t} = -\mu'_t$$

$$\Rightarrow \mu'_t = \rho \mu_t \Rightarrow \frac{d\mu_t}{\mu_t} = \rho dt \Rightarrow \ln \mu_t = \rho t + c_1,$$

$$\Rightarrow \mu_t = e^{\rho t + c_1} \quad \dots (ii)$$

$$\text{We know } W' = -c(t)$$

$$\text{So, } c_t = e^{-\rho t - c_1}$$

$$\Rightarrow \frac{dw}{dt} = -c_t \Rightarrow \int dw = -c_t dt$$

$$\Rightarrow w = -c_t t + c_2$$

Using boundary condition at time $T = t$ $w = 0$

$$\Rightarrow -c_t T + c_2 = 0 \Rightarrow c_2 = c_t T$$

$$\text{at } t = 0 \quad w = \phi \Rightarrow c_2 = \phi$$

$$\text{Setting } t = T \quad 0 = -e^{-\rho T - c_1} + \phi \Rightarrow \phi = e^{-\rho T - c_1}$$

$$\Rightarrow \ln \phi = -\rho T - c_1$$

$$\Rightarrow c_1 = \ln \phi + \rho T$$

$$\text{Hence } W^* = -e^{-\rho t - \ln \phi - \rho T} \cdot t + \phi$$

Q2. Consider an investor who at time $t = 0$ is endowed with initial capital of $x(0) = x_0 > 0$. At any time $t \in [0, T]$ where $T > 0$ is given he decides about his rate of consumption $c(t) \in [0, \bar{c}]$ where $\bar{c} > 0$ is a large maximum allowable rate of consumption thus his capital stock evolves according to $\dot{x} = \alpha x - c(t)$ where $\alpha > 0$ is a given rate of return the investor's time- t utility for consuming at rate $c(t)$ is $u(c(t))$ where $u: P_+ \rightarrow P$ is his increasing strictly concave utility function. The investor's problem is to find a consumption plan $c(t), t \in [0, T]$ so as to maximize his discounted utility.

$$J(c) = \int_0^T e^{-rt} u(c(t)) dt \quad u(ct) = \ln(ct)$$

$$x(0) = x_0 \quad x_T = 0$$

where $r \geq 0$ is a given discount rate subject to the solvency constraint that the capital stock $x(t)$ must be positive for all $t \in [0, T]$.

$$\text{Ans. } J(.) = \int_0^t e^{-rt} \ln(ct) dt \quad \text{s.t. } x' = \alpha - c(t)$$

$$\text{Let } H^* = g(.) + \mu_t h(.) \quad \text{where } g(.) = \ln c_t$$

$$h(.) = \alpha x - c$$

$$\text{So } \frac{\partial H}{\partial C} = 0 = \frac{1}{C} - \mu_t \Rightarrow C^* = \frac{1}{\mu(t)}$$

The costate equation (ref equation 23.40)

$$\mu'_t = r\mu_t - \frac{\partial g}{\partial x_t} - \mu_t \frac{\partial h}{\partial x_t} = r\mu_t - \alpha\mu_t$$

$$\Rightarrow \frac{d\mu_t}{dt} = (r - \alpha) \Rightarrow \mu^* = ke^{(r-\alpha)t} \text{ where } k \text{ is constant.}$$

From state equation $x' = \alpha x - c(t)$

$$\Rightarrow \frac{dx}{dt} - \alpha x = \frac{-1}{k e^{(r-\alpha)t}} = K_1 e^{(\alpha-r)t} \text{ where } K_1 \text{ is constant.}$$

Using the formula

$$x(t) = e^{-\int v dt} \left(A + \int w e^{\int v dt} dt \right)$$

$$\text{where } v = -\alpha \quad w = K_1 e^{(\alpha-r)t}$$

$$\begin{aligned} \text{we get } x(t) &= e^{-\int -\alpha dt} \left(A + \int K_1 e^{(\alpha-r)t} e^{\int -\alpha dt} dt \right) \\ &= e^{\alpha t} \left(A + \int K_1 e^{-rt} dt \right) \\ &= A e^{\alpha t} + \frac{-K}{r} e^{(\alpha-r)t} \end{aligned}$$

Using boundary condition $x(0) = x_0$

$$x_0 = A - \frac{K}{r}$$

$$x(T) = 0 \Rightarrow 0 = A e^{\alpha T} - \frac{K}{r} e^{(\alpha-r)T} = A e^{\alpha T} \left(A - \frac{K}{r} e^{-rT} \right) = 0$$

$$\text{as } \alpha > 0 \quad e^{\alpha T} > 0 \quad \text{so } A - \frac{K}{r} e^{-rT} = 0 \Rightarrow K = A r e^{rT}$$

$$\text{So } x_0 = A - \frac{A r e^{rT}}{r} = A(1 - e^{rT})$$

$$A = x_0(1 - e^{rT})^{-1} = \frac{x_0}{1 - e^{rT}}$$

$$\begin{aligned} x^*(t) &= \frac{x_0}{1 - e^{rT}} e^{\alpha t} - \frac{A r e^{rT}}{r} e^{(\alpha-r)t} \\ &= \frac{x_0}{1 - e^{rT}} e^{\alpha t} - \frac{x_0}{1 - e^{rT}} e^{(\alpha-r)t+rT} \end{aligned}$$

Q3. $\max_{0 \leq t \leq 1} \int_0^1 (x + u) dt \quad \text{s.t. } x' = 1 - u^2 \quad x(0) = 1 \quad x(1) \text{ is free.}$

Ans.

$$\max_{0 \leq t \leq 1} \int_0^1 (x + u) dt \quad \text{s.t. } x' = 1 - u^2 \quad x(0) = 1$$

From Hamiltonian $H(.) = x + u + \lambda(1 - u^2)$

necessary condition $\frac{\partial H}{\partial u} = 0 \Rightarrow 1 - 2\lambda u = 0$

$$\Rightarrow u^*(t) = \frac{1}{2}\lambda(t)$$

because $x(1)$ is free

we use boundary condition $\left. \frac{\partial s(\cdot)}{\partial x} - \lambda(t) \right|_{t=t_f} = 0$

as $s(\cdot) = 0$ (no terminal condition given) $\lambda(1) = 0$

Further $\frac{\partial H}{\partial x} = -\lambda'(t) \Rightarrow \lambda'(t) = -1$

$$\Rightarrow \frac{d\lambda}{dt} = -1 \Rightarrow d\lambda = -1dt$$

$$\Rightarrow \lambda(t) = -t + c_{\text{using}} \lambda(1) = 0$$

We get $\lambda(t) = 1 - t$

$$\text{So } u^* = \frac{1}{2}(1 - t)$$

Using the u^* in state dynamic equation (i.e., constraint)

$$x' = 1 - \frac{1}{4}(1 - t^2) \text{ given } (x(0) = 1)$$

$$\text{integrating we get } x^*(t) = t - \frac{1}{4}(1 - t) + \frac{5}{4} \text{ (Ans).}$$

Q4. Determine a control law that would drive the system from an initial state $x_1(0) = 0$, $x_2(0) = 2$ to final state $x_1(t_f) = x_2(t_f) = 0$ in minimum waste or minimal control effort

$$\text{P.I.} = \int_0^{t_f} \frac{1}{2} u^2(t) dt \quad \text{s.t.} \quad \dot{x}_1(t) = x_2(t)$$

$$\dot{x}_2(t) = -x_1(t) + u(t)$$

here $S = (x_f, t_f) = 0$

Ans. Rewriting the problem P.I. Performance index

$$= \int_0^{t_f} \frac{1}{2} u^2 dt \quad \dots (1)$$

$$\text{s.t. } \dot{x}_1(t) = x_2(t) \quad \dots (2)$$

$$\dot{x}_2(t) = -x_1(t) + u(t) \quad \dots (3)$$

$$\text{given } x_1(0), x_2(0) = 2 \quad x_1 t_f = x_2 t_f = 0$$

forming Hamiltonian

$$H(.) = \frac{1}{2}u^2 + \lambda_1 x_2 - \lambda_2 x_1 + \lambda_2 u \quad \dots (4)$$

For necessary condition $\frac{\partial H(.)}{\partial u} = 0$

$$\Rightarrow \frac{1}{2}.2u. + \lambda_2 \Rightarrow u = -\lambda_2 \text{ or more precisely } u^*(t) = -\lambda_2(t) \quad \dots (5)$$

Using (5) in equation (3)

$$\dot{x}_2(t) = -x_1(t) - \lambda_2(t) \quad \dots (6)$$

Co-state equation $\frac{\partial H(.)}{\partial x_1(t)} = -\dot{\lambda}_1(t) \quad \dots (7)$

$$\frac{\partial H(.)}{\partial x_2(t)} = -\dot{\lambda}_2(t) \quad \dots (8)$$

From eq (7) $-\lambda_2(t) = -\dot{\lambda}_1(t) \Rightarrow \lambda_2(t) = \dot{\lambda}_1(t) \quad \dots (9)$

From eq (8) $\lambda_1(t) = -\dot{\lambda}_2(t) \quad \dots (10)$

From eq (9) and (10) $\ddot{\lambda}_2(t) = -\lambda_2(t)$

So that 2nd order homogeneous equation

$$\begin{aligned} \lambda_2(t) &= \alpha_1 e^{jt} + \alpha_2 e^{-jt} \quad \text{where } j = \sqrt{-1} \\ &= (\alpha_1 + \alpha_2) \cos t + j(\alpha_1 - \alpha_2) \sin t \quad \dots (11) \end{aligned}$$

at time $t = 0$ $\lambda_2(0) = \alpha_1 + \alpha_2$ (Note α_1 and α_2 are constants)

note $\dot{\lambda}_2(t) = -(\alpha_1 + \alpha_2) \sin t + j(\alpha_1 - \alpha_2) \cos t$

at time $t = 0$ $\dot{\lambda}_2(0) = j(\alpha_1 - \alpha_2) \quad \dots (12)$

as we know $\dot{\lambda}_2(t) = -\lambda_1(t)$

so $-\lambda_1(0) = j(\alpha_1 - \alpha_2) \quad \dots (13)$

so we can write e.g. (11) $\lambda_2(t) = \lambda_2(0) \cos t - \lambda_1(0) \sin t \quad \dots (14)$

$$\dot{\lambda}_2(t) = \lambda_2(0) - \sin t - \lambda_1(0) \cos t$$

Using eq. (10) $-\lambda_1(t) = \lambda_2(0)(-\sin t) - \lambda_1(0) \cos t \quad \dots (15)$

From eq. (2) $\ddot{x}_1(t) = \dot{x}_2(t) = -x_1(t) + u_t$

and (3) as $u^*(t) = -\lambda_2(t)$

$$\ddot{x}_1(t) = -x_1(t) - [\lambda_2(0) \cos t - \lambda_1(0) \sin t]$$

(using eq. 14)

The function within [-] can be thought of as input to the system or forcing function to differential equation.

Therefore, $x_1(t) = 2 \sin t + \frac{1}{2} \lambda_1(0) [\sin t - \cos t] - \frac{1}{2} \lambda_2(0) t \sin t \quad \dots (16)$

From eq. (2) $\dot{x}_1 = x_2(t)$

$$= 2 \cos t + \frac{1}{2} \lambda_1(0) [\cos t - (\cos t + t(-\sin t))] - \frac{1}{2} \lambda_2(0) (\sin t + t \cos t)$$

Therefore, $x_2(t) = 2 \cos t + \frac{1}{2} \lambda_1(0) (t \sin t) - \frac{1}{2} \lambda_2(0) (\sin t + t \cos t) \dots (17)$

applying the given end-point boundary condition we

get at $t = t_f$ $x_1(t_f) = x_2(t_f) = 0$

here t_f as free $x(t_f)$ is fixed

$$\left[H(.) + \frac{\partial s(x(t), t)}{\partial t} \right]_{t=t_f} \delta t_f = 0$$

as $\delta t_f \neq 0$ and $s(x(t), t) = 0$ (no terminal condition given in problem)

$$H(.)|_{t=t_f} = 0$$

Rewriting equation (4)

$$H(.) = -\frac{1}{2} \lambda_2^2(t_f) + \lambda_1(t_f) x_2(t_f) - \lambda_2(t_f) \cdot (x_1 t_f) = 0$$

Using initial condition $x_1 t_f = x_2 t_f = 0$

$$\lambda_2(t_f) = 0$$

From e.g. (14) $\lambda_2(t_f) = \lambda_2(0) \cos t_f - \lambda_1(0) \sin t_f$

$$\Rightarrow 0 = \lambda_2(0) \cos t_f - \lambda_1(0) \sin t_f \quad \dots (18)$$

We have 3 equations e.g. (16), (17) and (18) [put $t = t_f$ in (16) and (17)] and 18 is given. We have 3 unknowns $\lambda_2(0), \lambda_1(0)$ and t_f these are non-linear equations need not to be solved analytically. Rather to be done by numerical method.

UNIT 24 ECONOMIC APPLICATIONS OF DYNAMIC OPTIMISATION

Structure

- 24.0 Objectives
- 24.1 Introduction
- 24.2 Discounting
- 24.3 Calculation of Current Value (present value) of Investment Streams
- 24.4 Optimal Rate of Extraction of Exhaustible Resources by Monopoly
- 24.5 Optimal Control of a Generic Renewable Resource
- 24.6 Optimal Depletion of Non-Renewable Resource
- 24.7 Neoclassical Investment Theory
- 24.8 Optimal Growth Model
- 24.9 Neoclassical Theory of Investment with Adjustment Costs
- 24.10 Let Us Sum Up
- 24.11 Key Words
- 24.12 Answers/Hints to Check Your Progress Exercises
- 24.13 Exercises

24.0 OBJECTIVES

After going through this unit, you should be able to

- compute the present value of economic variables;
- appreciate the mathematical tool of Hamiltonian functional used in environment economics analysis;
- derive optimisation criteria in few standard macroeconomic models; and
- perform analysis of dynamic models post optimality condition.

24.1 INTRODUCTION

In this concluding unit of Block 7, by using the formula derived as well as explained in Unit 23, we exemplify the beauty of dynamic optimisation. Therefore, a grasp of contents of Unit 23 is prerequisite for reading Unit-24. Sourced from established literature, the Unit 24 presents six applications-3 each from microeconomics and macroeconomics, which demonstrate model formulation, optimal solution & necessary analysis. The concept of dynamic optimisation is well founded and its application in economics is ever expanding. Unit 24 is a humble yet solid endeavour to innervate students for exploring the vast area of applied dynamic optimisation in economics.

24.2 DISCOUNTING

Perhaps we all have heard the maxim – Money received today is more than money received tomorrow; and this message is true in the world even without inflation. The question therefore begs, why that-discounting happens. The answer lies with the fact, that money is attached with interest it can accrue. Consider the case of simple interest for one year being 10%. If we invest ₹ 1 today, we will get ₹ 1.1 after a year from now. In other words, one can be indifferent between ₹ 1 today and ₹ 1.1 a year after one year from today, given market interest rate 10%. If we want to have ₹ 1 a year from Now, we need to invest $1/(1+0.1) \cong 0.91$ ₹ today. Put another way having to wait a year to receive that ₹ 1 make it worth ₹ 0.09 less to us today.

The value today of payment to be received in the future is called present value. We use present values to compare cost and benefits that occur at different points in time.

Similarly, considering interest-rate 10% per year, if we receive ₹ 0.83 today that would be equal to receiving ₹ 1 after 2 years from now.

The formula for compounding is

$$y = P \left(1 + \frac{r}{100} \right)^T$$

where p = principal or money today

r = rate of interest

T = Time period of consideration

Y = net money after compounding at end of ‘ T ’ year.

The present value PV of x represents how much you need to invest today in order to have x at end of T years from now. Symbolically,

$$PV(x) = \left(\frac{1}{1+r} \right)^T x = \frac{x}{(1+r)^T} \quad \dots (24.1)$$

where T is the number of the year from today in future, in which we receive x .

If x is a cost that occurs in the future, you can think of the present value as money you need to set aside today in order to cover that cost the day it takes place. In generalized case ‘ r ’ the annual rate of interest can be calculated monthly, daily etc. Similarly, T can be one year, half year, quarter year etc. Our reference year does not need to be the present day; it could also be next year or 10 years from now, what is important is that all costs and benefits be valued relative to the same reference year, for the sake of analysis.

24.3 CALCULATION OF CURRENT VALUE OF INVESTMENT STREAMS

The value of stream of cost and benefits that occur at different periods in time in future, is the sum of the discounted individual values. The present value of an investment stream is

$$PV = x_0 + \frac{x_1}{1+r} + \frac{x_2}{(1+r)^2} + \frac{x_3}{(1+r)^3} + \dots \quad (24.2)$$

for all costs and benefits x^t that occur at time t in future. If same amount $x_t = x$ is received during each period until the end to time (perpetuity) then, it forms a geometric series with $y = \frac{1}{1+r} < 1$

[Recollect the formula of summation of ~~for~~ an infinite geometric series with common ratio $y < 1$:

$$1 + y + y^2 + y^3 + \dots = \frac{1}{1-y}$$

We can simplify equation (24.2)

$$\begin{aligned} PV &= x \left(1 + \left(\frac{1}{1+r} \right) + \left(\frac{1}{1+r} \right)^2 + \left(\frac{1}{1+r} \right)^3 + \dots \right) \\ &= x \left(\frac{1}{1 - \left(\frac{1}{1+r} \right)} \right) = x \left(\frac{1}{\frac{1+r-1}{1+r}} \right) = x \left(\frac{1+r}{r} \right) \end{aligned} \quad \dots (24.3)$$

If there is no payment in the initial period, i.e.,

if the investment stream is PV in equation (24.3) above minus x

$$\begin{aligned} \text{Then new PV will be } & \frac{x}{1+r} + \frac{x}{(1+r)^2} \dots \\ & = \frac{x(1+r)}{r} - x = x \left(\frac{1+r-r}{r} \right) = \frac{x}{r} \end{aligned} \quad \dots (24.4)$$

24.4 OPTIMAL RATE OF EXTRACTION OF EXHAUSTIBLE RESOURCES BY MONOPOLY¹

Let the control problem be

¹ Monopoly and the rate of extraction of exhaustible Resources, Stiglitz, J (1976), The American Economic Review, Vol. 66, No. 4 PP. 655-661.

$$\max \int_0^T [p_t q_t - c(q_t)] e^{-rt} dt \quad \dots (24.5)$$

$$\text{Subject to } x' = -q_t \quad \dots (24.6)$$

where p_t = market price of extracted resource at period, t

q_t = Quantity of resources sold/extracted at period

t (This has an assumption i.e., no resources is stored for next period at source or zero inventory)

$c(q_t)$ = cost of resource extraction at period t

x' = rate of change of stock of resource

e^{-rt} = discounting factor.

(Please refer to the discounted problem in unit: 23)

We can write $H^* = p_t q_t - c(q_t) - \mu_t q_t$

$$\text{Setting } \frac{\partial H}{\partial q_t} = 0 \Rightarrow p_t + q_t \frac{dp_t}{dq_t} - c'(q_t) - \mu_t = 0 \quad \dots (24.7)$$

Total revenue = $TR = pq$

Hence marginal revenue $R'(q) = p_t + q_t \frac{dp}{dq}$

From necessary condition of optimisation

$$\mu' = r \mu_t$$

From equation (24.7) $\mu_t = \left(p_t + q_t \frac{dp}{dq} \right) - c'(q)$

Special case elasticity $\epsilon = \text{constant}$ and $c'(q) = 0$

$$R' = \left(p + q \frac{dp}{dq} \right) = p \left(1 + \frac{dp}{dq} \cdot \frac{q}{p} \right) = p \left(1 + \frac{1}{\epsilon} \right)$$

$$\Rightarrow R' = kp \text{ where } k = \text{constant} = \left(1 + \frac{1}{\epsilon} \right)$$

We can write $\mu_t = R' - c'(q)$ if $c'(q) = 0$ the $\mu_t = R'$

$$\mu' = \frac{d(R')}{dt} = rR'$$

If we take $\frac{dp}{dt} = rp$ then $\frac{d(kp)}{dt} = rkp$

If $\frac{dc'}{dt} = 0$ then $\frac{d(R' - c')}{dt} = \frac{dR'}{dt} = r(R' - c')$

So we can write